

AI-Native Network Digital Twin for Intelligent Network Management in 6G

Wen Wu , Xinyu Huang , Meng Qin , Qihao Li , Nan Cheng , and Tom H. Luan 

ABSTRACT

As a pivotal virtualization technology, the network digital twin (DT) is expected to accurately reflect real-time status and abstract features in the ongoing sixth generation (6G) networks. In this article, we propose an artificial intelligence (AI)-native network DT framework for 6G networks to enable the synergy of AI and network DT, thereby facilitating intelligent network management. In the proposed framework, AI models are utilized to establish network DT models to support network status prediction, network pattern abstraction, and network management decision-making. Furthermore, potential solutions are proposed to enhance the performance of network DT. Finally, a case study is presented, followed by a discussion of open research issues that are essential for the AI-native network DT in 6G networks.

INTRODUCTION

With the rollout of the fifth-generation (5G) networks worldwide, industry and academia have begun exploring the next generation of wireless networks, namely 6G networks. As the successor to 5G, 6G is anticipated to support an unprecedented variety of services with significantly increased service demands and evolve towards intelligent networking [1], [2]. Artificial intelligence (AI) is expected to penetrate every corner of wireless networks, including end users, the network edge, and the cloud, leading to ubiquitous network intelligence, i.e., AI-native networks. This intelligence, which spreads across network-wide entities, will enable intelligent network management for the evolving 6G networks.

Digital twin (DT) is a promising technology for advancing network management in 6G networks [3]. The concept of DT originates from the manufacturing industry [4], which constructs detailed digital replicas of physical systems to simulate, monitor, and optimize these systems in real time, thereby providing a significant leap in operational efficiency. This concept has progressively been applied to wireless networks, i.e., *network DT*. The network DT enhances the performance of traditional network management by integrating three key components: user DT, infrastructure DT, and slice DT. User DTs capture and model individual user behavior patterns, thereby supporting the analysis of personalized service demands for

network optimization. Infrastructure DTs characterize the operation status of the network from the perspective of network infrastructure, such as base stations, edge servers, and cloud servers, thereby analyzing network traffic patterns, balancing network workloads, and validating network management policies. Slice DTs facilitate accurate and flexible resource provisioning by estimating service demands from historical spatio-temporal data.

However, achieving AI-native network DT faces several challenges. Firstly, collecting vast amounts of data for network DT construction puts substantial pressure on communication networks. Developing efficient data transmission techniques is essential to mitigate data collection overhead. Secondly, the spatial-temporal dynamics of communication networks necessitate a scalable network DT model for low-energy data processing. It is crucial to create an inference mechanism for a scalable network DT model to enhance energy efficiency. Thirdly, the network DT may become increasingly inaccurate if the models cannot adapt to inherent data drift in the network, triggered by shifts in user behavior, device interoperability, and service demands. It is essential to develop a method for updating the adaptive network DT model to improve network DT fidelity.

In this article, we propose an AI-native network DT framework to facilitate intelligent network management in 6G. AI models are collaboratively embedded in user, infrastructure, and slice DTs to achieve AI-native network status emulation, network feature abstraction, and network decision-making. To address the above challenges, we first adopt advanced semantic communication techniques to realize efficient data collection. Then, we develop an adaptive network DT model inference scheme that reduces energy consumption. Next, we propose a dual error-based model update mechanism to trigger network DT model updates adaptively in an online manner. A case study related to AI-native adaptive user DT model inference and update is presented, followed by a discussion on research issues.

The remainder of this article is organized as follows. In the section “*Network DT and AI Technology*,” network DT and advanced AI techniques are discussed, followed by the proposed network DT framework in the section “*AI-Native Network*

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DT Framework for Intelligent Network Management.” Then, we discuss the challenges and potential solutions in the section “Challenges and Solutions.” Next, a case study and open research issues are presented in sections “Case Study” and “Open Research Issues,” respectively. Finally, we conclude this article in the section “Conclusion.”

NETWORK DT AND AI TECHNOLOGY

NETWORK DT

A network DT is a virtual representation that accurately replicates and simulates the structure, operations, and dynamics of a physical communication network. As shown in Fig. 1, depending on the types of virtualized physical network entities, the network DT is categorized into three groups: user DT, infrastructure DT, and slice DT, which are detailed below.

User DT: A user DT corresponds to an end user, which characterizes fine-grained user status. The user status data can be classified into three types: profile information, behavior information, and networking information. Deploying at network edge, user DTs help local network controllers perceive user status in real time. As such, user DTs are able to consolidate behavior and networking information for efficient correlation analysis, thereby enabling intelligent network management.

Infrastructure DT: An infrastructure DT is a digital mirror of a network infrastructure that reflects its network status, configuration, and performance information. This information can be obtained from real-time reports of network infrastructures or prediction models. Infrastructure DTs are typically deployed at network edge nodes, which help local network controllers perceive real-time network infrastructure status, enabling efficient communication, sensing, and computing collaboration. The infrastructure DT consists of extensive network data and sophisticated modeling techniques to optimize network management. Detailed topology and configuration information, along with performance metrics, are collected and integrated from the network infrastructure. The data is processed through advanced methods, including data transformation and integration, which consolidate disparate data into a cohesive framework, and traffic pattern analysis, which alleviates network overloads in certain infrastructures.

Slice DT: A slice DT corresponds to a network slice, which is an isolated logical network in a partitioned physical network that manages virtualized resources to meet various service requirements [5]. A slice DT usually stores data related to service requirements, user group information, configuration details, and performance data, which can be abstracted from the user DT and infrastructure DT. Slice DTs are typically deployed at network core nodes, assisting the global network controller in efficiently slicing the entire network resources. The slice DT comprises two primary components: data and models. The data component includes detailed service requirements, such as delay and priority, as well as user group information like mobility patterns and resource demands. Slicing configuration and performance data is also stored in the slice DT, including slice type, resource reservation, and

quality of service (QoS) violation metrics. In terms of models, the slice DT leverages data transformation and integration to ensure consistency across various sources. The slice DT employs user satisfaction analysis to evaluate slicing performance and adjust dynamic slicing policies. Furthermore, policy validation ensures that operational policies align with compliance standards and performance objectives, while predictive models facilitate proactive network resource reservation [6].

The network DTs are organized in a hierarchical structure, with user DTs at the bottom layer, infrastructure DTs in the middle layer, and slice DTs at the top layer. User DTs emulate user status and abstract features, which are aggregated by infrastructure DTs to analyze regional traffic patterns for local network management. The data from infrastructure DTs is further aggregated by slice DTs to provide some insights, such as spatio-temporal resource demand distributions, which allows the global controller to make adaptive network slicing decisions. It is worth noting that the capabilities of network DTs are inherently compatible with the key features envisioned for 6G networks. 6G’s ultra-low latency and high reliability support real-time synchronization between the physical and virtual environments, which is essential for precise status emulation and timely control. In addition, the pervasive AI in 6G enables the deployment of intelligent DT functions across edge and cloud servers to construct hierarchical network DTs.

AI TECHNOLOGY

AI technologies play a crucial role in enhancing the capabilities of DTs for future 6G network management, particularly in network status prediction, feature extraction, and decision-making. These capabilities are primarily bolstered through three distinct categories of AI techniques: foundation AI, decision-making AI, and analytical AI, as shown in Table 1.

1) Foundation AI: Foundation AI, represented by large-scale models such as generative pre-training transformer (GPT) and bidirectional encoder representations from transformers (BERT), significantly enhances the network DT’s ability to comprehend and interpret complex network data across diverse domains. Due to their generalized and cross-domain capabilities, these models effectively facilitate knowledge reasoning and multi-modal perception [7], [8]. They are particularly adept at extracting high-level network features and understanding intricate network states by integrating vast amounts of heterogeneous data. However, their large parameter sizes require substantial computational and memory resources, which typically leads to deployment in cloud-based or centralized infrastructures.

2) Decision-Making AI: Decision-making AI, such as time-series forecasting and reinforcement learning, can enhance the predictive and automated decision-making capabilities of network DTs. Temporal forecasting algorithms, such as lightweight transformers or long short-term memory (LSTM), enable network DTs to accurately predict network status, including traffic loads, latency, and QoS metrics [9]. Reinforcement learning algorithms further facilitate adaptive, dynamic, and uncertainty-aware decisions. These

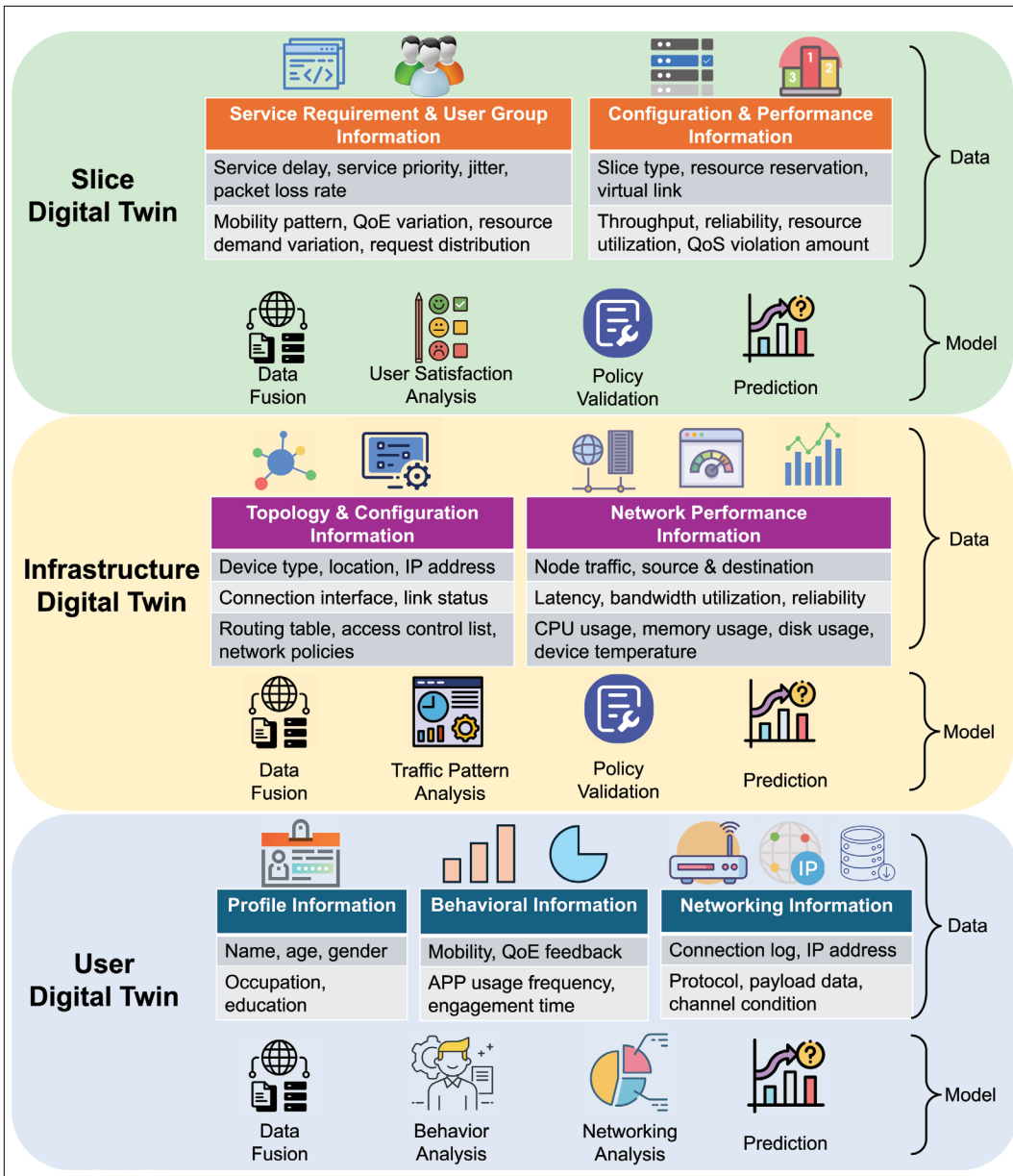


FIGURE 1. The three categories of network DT, i.e., user, infrastructure, and slice DTs.

techniques, which require relatively lower memory and computational resources, are ideal for deployment across edge computing nodes, thereby enabling real-time and proactive network management.

3) Analytical AI: Analytical AI, including autoencoders and graph neural networks (GNNs), primarily enhances the anomaly diagnosis and feature abstraction capabilities of network DTs. Their inherent sensitivity to anomalies, structural awareness, and explainability facilitate efficient root-cause analysis and reliable fault detection within network operations [10]. Explainable AI further aids in transparent decision-making, enabling network operators to trust and swiftly act upon AI-generated insights. With their low memory and modest computational demands, these models are well-suited for real-time or near-real-time deployments at network edges to support continuous monitoring and efficient diagnostic analytics. By effectively integrating these categories of AI

Category	Foundation AI	Decision-making AI	Analytical AI
Typical Models	- GPT - BERT	- Reinforcement learning - Time-series forecasting	- Autoencoder - Graph neural network
Characteristics	- Generalized - Cross-domain - Large parameter size	- Temporal awareness - Adaptive & dynamic - Uncertainty awareness	- Anomaly-sensitive - Explainable - Structure-aware
Functions	- Knowledge reasoning - Multi-modal perception	- Status prediction - Automated decision-making	- Anomaly diagnosis - Feature abstraction
Memory	High	Low to medium	Low
Computation	High	Low	Low to medium

TABLE 1. The comparison of AI technologies for network DT.

technology, network DTs can achieve automated intelligence regarding predictive accuracy, operational efficiency, and reliability in 6G network management.

AI-NATIVE NETWORK DT FRAMEWORK FOR INTELLIGENT NETWORK MANAGEMENT

FRAMEWORK OVERVIEW

As shown in Fig. 2, we propose an overall AI-native network DT framework for intelligent network management, which leverages the advantages of AI technology in data generation, data analysis, and decision-making. The proposed framework comprises the physical network, user DT, infrastructure DT, and slice DT. The physical network is a multi-tier structure, where the centralized controller manages large-timescale network slicing and the local controller oversees real-time network operations. Data from user DTs are initially abstracted into the infrastructure DT, where they are combined with infrastructure status to facilitate real-time network operations. Subsequently, data from both the user DT and infrastructure DT is aggregated into the slice DT to update slice status and refine network slicing strategies. In the following sections, we will demonstrate how to leverage the framework to facilitate intelligent network management.

1) AI-Native User DT: Since an essential function of the user DT is to characterize real-time user status on the network side, frequently collecting data from the physical network not only incurs high costs but also poses risks of communication interruptions due to user mobility. Recurrent neural networks (RNNs), known for their proficiency in analyzing sequential and time-series data, are pivotal in continuously updating user status by capturing temporal patterns. By deploying lightweight RNN models on edge servers, user status in the user DT can be timely updated through data generation based on historical data. Another critical function of the user DT is abstracting

users' traffic and behavior patterns. Convolutional neural networks (CNNs) can effectively analyze network conditions to extract complex spatial and temporal patterns. Furthermore, autoencoders streamline the complexity of the user DT by distilling high-dimensional data into its most salient features through the encoding-decoding mechanism. This process transforms the intricate user DT data into useful features, such as behavior patterns, thereby boosting network management efficiency. For instance, user swipe probability distributions in short video streaming services can be utilized to estimate spectrum resource demands in the communication resource reservation stage, which can reduce resource wastage. User request distributions can be used to analyze resource demands in the computing resource reservation stage, which can improve computing resource utilization. Together, these AI models can help the user DT monitor real-time user status, abstract behavior patterns, and compress high-dimensional user data into low-dimensional representations.

2) AI-Native Infrastructure DT: As shown in Fig. 2, GNN, CNNs, deep reinforcement learning (DRL) models, and network simulators such as NS-3 are integrated into the infrastructure DT to enhance the efficiency of construction and maintenance. GNNs are utilized to analyze the network's topology and configuration, excelling in managing the correlated data inherent in networks, such as the connections between network infrastructures and the transmission of signals or data through these connections [11]. CNNs are adept at recognizing patterns from multidimensional data, including traffic flow, bandwidth usage, and anomaly detection. DRL employs an agent-based approach where each agent interacts with a simulated network environment provided by the network simulator, optimizing network operations through continuous learning from actions and

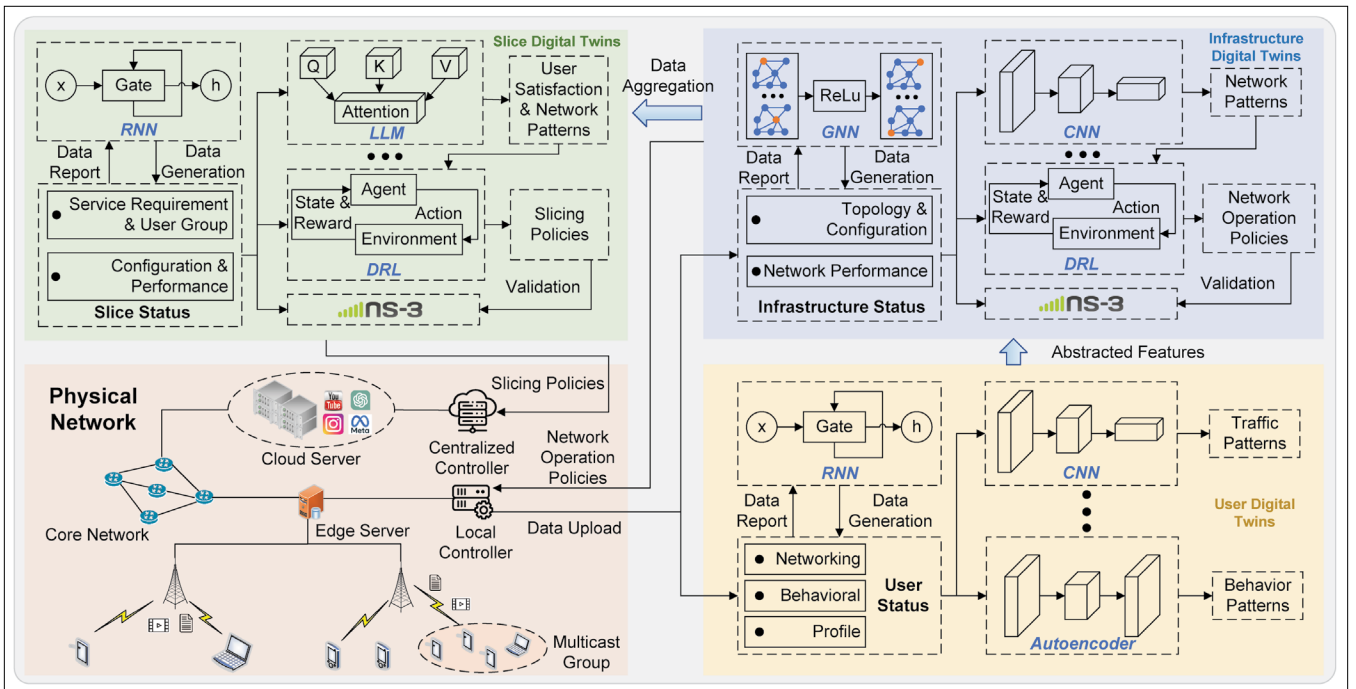


FIGURE 2. The proposed AI-native network DT framework for intelligent network management.

received rewards. The network simulator serves as a pivotal tool in the infrastructure DT, providing a realistic simulation environment for evaluating network policies and ensuring that the infrastructure DT accurately captures the dynamics of the physical network. Together, these technologies create an efficient and robust infrastructure DT that can provide the network controller with holistic infrastructure status, distilled network patterns, and autonomously optimized network decisions.

3) AI-Native Slice DT: The integration of RNNs, large language model (LLM), DRL, and the network simulator is an effective approach to realize slice DT-based intelligent network management. RNNs are utilized for generating and processing sequential data within the slice DT, particularly useful for analyzing time-series data related to network performance. The LLM equipped with attention mechanism, is adept at processing and understanding large volumes of textual data. In the context of slice DTs, the LLM can analyze user interaction and network traffic data to obtain user satisfaction and network patterns, respectively. The attention mechanism allows for the prioritization of relevant information, making the data

analysis more focused and efficient. DRL utilizes sophisticated algorithms where an agent interacts with a simulated network environment to learn and refine slicing policies through trial and error [12]. This continuous interaction, supported by simulation data from the network simulator, helps in developing robust and efficient network slicing strategies. Overall, AI-native slice DT provides effective intelligent network management through data prediction, textual data analysis, and iterative policy learning in a simulated environment.

WORKFLOW

As shown in Fig. 3, we present a detailed workflow within the proposed framework. At the outset, user and infrastructure DTs employ RNN models to predict future user and network status. If the discrepancy between predicted and actual data exceeds a predefined threshold, user and infrastructure DTs adjust the data collection frequencies from access points (APs) to enhance data freshness and accuracy. Subsequently, user DTs utilize CNN and autoencoder models to extract user features, which are then reported to infrastructure DTs. Within infrastructure DTs, CNN

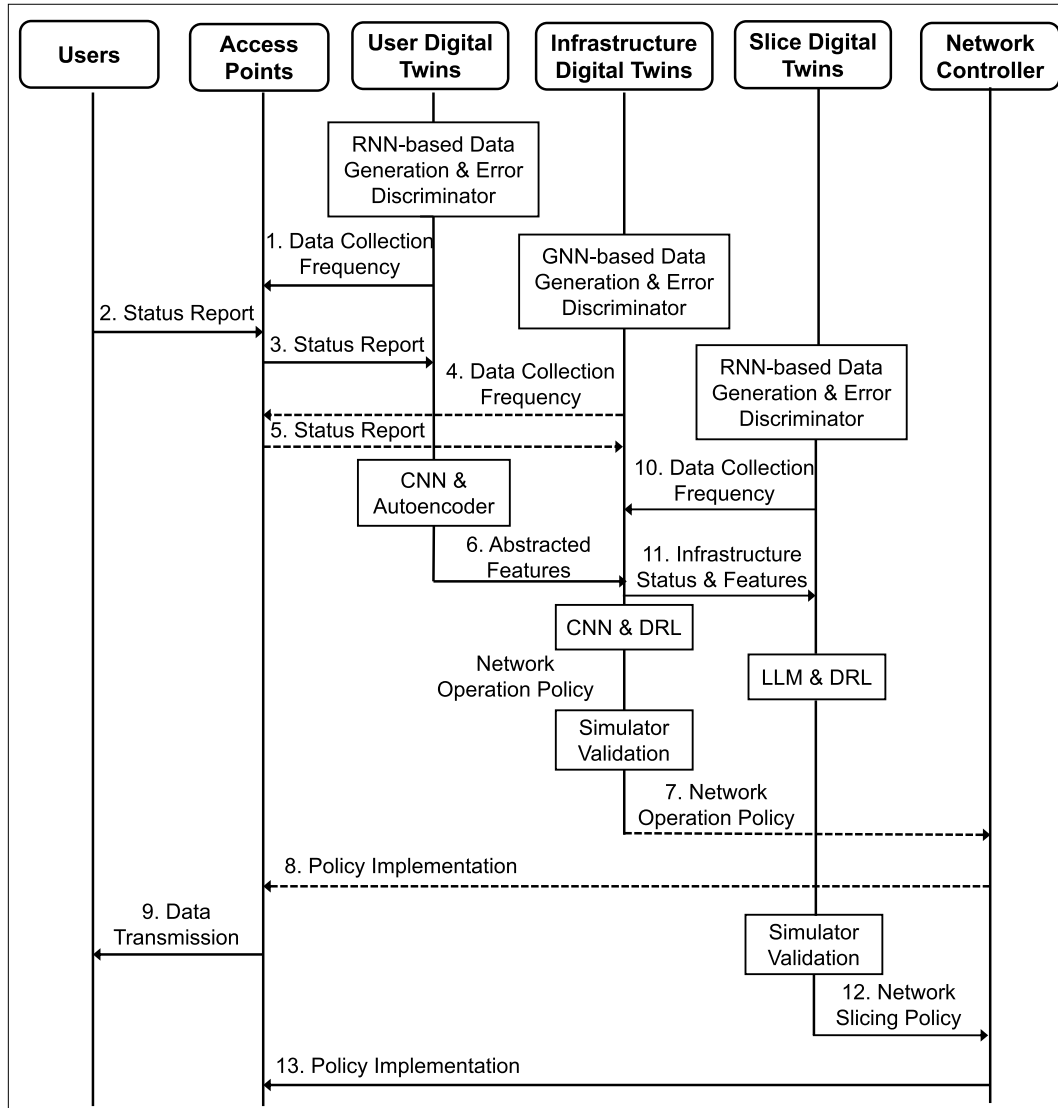


FIGURE 3. Detailed workflow of the proposed AI-native network DT framework.

models analyze network patterns, feeding into DRL models to create network operation policies, which are validated in an internal infrastructure DT simulator to ensure network robustness. The processed data from user and infrastructure DTs is escalated to the slice DT, where LLM and DRL models analyze network patterns and user satisfaction to inform network slicing decisions. These decisions undergo validation in an internal slice DT simulator to confirm their effectiveness and reliability. Ultimately, the infrastructure and slice DTs forward these refined network operation and slicing decisions to the network controller for implementation. To ensure seamless collaboration among heterogeneous AI models, the proposed framework adopts a modular design with hierarchical data flows, asynchronous inference scheduling, and standardized feature interfaces, thereby enabling efficient and consistent end-to-end decision-making. The operating cycles of user and infrastructure DTs run at a small timescale (e.g., sub-second to seconds), which enable rapid local updates. In contrast, slice DTs operate at a long timescale (e.g., tens of minutes or hours), which aggregate abstracted features from the edge nodes for strategic resource planning.

CHALLENGES AND SOLUTIONS

CHALLENGES

1) Efficient Data Collection: Constructing network DTs typically requires the collection of massive multimodal data, placing significant traffic pressure on mobile communication networks. Multimodal data includes various types of information, such as video, audio, and network logs, each necessitating different handling, storage, and transmission protocols. The volume of data generated by IoT devices and high-bandwidth applications can be immense. Managing this influx of data without causing network congestion and maintaining low packet loss poses a significant challenge.

2) Scalable Data Processing: Operating and maintaining a network DT with low energy consumption is a significant challenge, primarily due to the need to balance computational intensity with the accuracy of DT data processing. A network DT requires continuous and real-time data processing to accurately mirror the physical network. This necessitates substantial computational resources, which inherently increases energy demands. Furthermore, the requirement for high-fidelity simulations to predict network behavior under various scenarios further escalates energy needs. Each simulation can involve intensive computational tasks, including data analysis, model updating, and synchronization across multiple DT components. Therefore, it is essential to design an efficient data processing mechanism for the DT to achieve energy efficiency.

3) Adaptive Model Update: Constructing and maintaining a high-fidelity network DT, which encompasses user, infrastructure, and slice DTs, presents significant challenges due to the obsolescence of AI models caused by data drift. Specifically, a network DT typically utilizes a variety of AI models to perform real-time mappings and extract salient network features. However, as network dynamics evolve, changes in user

behaviors, device interoperability, and service demands inherently cause data characteristics to drift. This drift can significantly degrade the performance of existing AI models, rendering them obsolete. Therefore, timely recalibration of network DT models is critical to align with network dynamics.

POTENTIAL SOLUTIONS

1) Advanced Semantic Communication Techniques: To tackle the challenges associated with the substantial overhead of DT data collection in networks, advanced semantic communication techniques present a promising solution. These techniques prioritize the semantic aspect of the transmitted data, emphasizing the meaning and relevance of the information rather than simply the raw data. Typically, semantic communication methods employ AI-driven algorithms to encode and decode data based on its contextual significance, facilitating the prioritization of critical information that directly influences the fidelity of the network DT. By delivering deep semantic data to decision-making processes, these techniques can reduce bandwidth requirements and ease the burden on network resources. Moreover, semantic communication techniques can dynamically adapt to changes in network conditions, ensuring that the most relevant data is processed and transmitted first, thereby maintaining the integrity and responsiveness of the network DT.

2) Adaptive Network DT Model Inference: To operate and maintain a network DT with energy efficiency, an adaptive DT model inference scheme is essential. Specifically, this scheme involves using lightweight AI algorithms for simpler, common networking tasks such as bandwidth allocation and QoS monitoring, where minimal computational resources are adequate. For more complex and critical tasks like dynamic routing optimization and network anomaly detection, which require deeper analysis and predictive capabilities, more powerful and computationally intensive AI models are employed. This adaptive approach enables the network DT to scale its computational needs based on the complexity and importance of the task, ensuring that the network remains efficient and responsive without excessive energy consumption. Additionally, network DT models can utilize neural networks with varying depths to process data. When network dynamics are low, lightweight models are activated to perform inference with satisfactory efficiency. Under high dynamics, more complex models are utilized to ensure performance. A flexible triggering mechanism can be designed to dynamically select appropriate models based on real-time network conditions. Rather than relying on static thresholds, the inference process can be orchestrated using a context-aware scheduling policy that considers traffic patterns, device energy budgets, service urgency, and prediction uncertainty. By formulating this trade-off as a multi-objective optimization or reinforcement learning problem, the system can dynamically balance energy efficiency with quality of experience (QoE) performance in real time. Therefore, a versatile and scalable AI model inference scheme should be developed to support an energy-efficient network DT.

3) Dual Error-Based Model Update Mechanism: To effectively counter the challenge posed by AI model obsolescence in the network DT, we propose a dual error-based model update mechanism. For data with accessible true labels, the mean squared error metric quantifies discrepancies between predicted outcomes and actual labels. Conversely, for data where true labels are unattainable, such as user or network features, we select the entropy of the model's output probability distribution to measure the error. This dual-error assessment framework facilitates a flexible model update. The network DT model updates are executed through incremental learning, allowing model recalibration without the burden of retraining from scratch [13]. The proposed error-based model update mechanism ensures that the network DT remains robust and accurate in dynamic network environments.

CASE STUDY

CONSIDERED SCENARIO

An adaptive user DT-assisted multicast short video streaming (MSVS) scenario is considered, which consists of multiple base stations, users with multicast groups (MGs), one edge server equipped with a transcoder for video transcoding, multiple user DTs, and a network controller. Users within one base station are clustered into multiple MGs, where the same short video streaming is transmitted to one MG through multicast transmission. The network operates in a time-slotted manner. In each real-time scheduling slot, the user DT can emulate user status and abstract user behavior patterns, which is leveraged to make customized communication, computing, and control (3C) management to improve user QoE, including multicast rebuffer time, video quality, and quality variation. The constructed user DT consists of user status data and AI models, which is detailed as follows.

- **DT Data Consists of Two Types of Information:** networking-related data and behavior-related data, which are used to characterize user status. The former pertains to users' channel conditions for estimating their transmission capabilities. The latter encompasses users' locations, swipe timestamps, and preferences, which help analyze users' mobility trajectories, swipe probability distributions, and video popularity. User DT data originates from two aspects, i.e., realistic data collected from the physical network and predicted data from the LSTM model. The real-world video streaming dataset¹ and `propagationModel` at Matlab are utilized to simulate data collection process from the physical network.
- **DT Models:** Two types of models, namely the status prediction model and the feature abstraction model, are used to analyze user status. Firstly, based on the data collected from the physical network, the user DT conducts status prediction by using the LSTM model to reduce data collection costs. Secondly, the designed autoencoder-based double deep Q network (DDQN) model clusters users and

abstracts users' behavior patterns, including swipe probability distributions and recommended video lists. Specifically, the autoencoder compresses time-series user DT data into low-dimensional latent features. The DDQN analyzes the latent features to determine the appropriate number of clusters. Based on the compressed user DT data and the number of clusters, the K-means++ method achieves rapid user clustering. The statistics and estimation module outputs the swipe probability distribution and recommended video list for each MG. User DT models are updated through incremental learning [14].

The adaptive user DT-assisted MSVS operates as follows. In the network operation stage, an appropriate user DT feature abstraction model is first activated to conduct model inference for MG update. Then, video transcoding and multicast transmission begin based on generative AI-driven 3C management to optimize user QoE. A cumulative error detector monitors the accuracy of predicted user status and clustering results. If the cumulative error surpasses a specified threshold, the network transitions to the user DT model update stage. In this stage, the user DT model is updated concurrently with video transcoding and multicast transmission. Once the user DT model update is completed, the network returns to the network operation stage.

SIMULATION RESULTS

To evaluate the influence of adaptive user DT model inference and updates on QoE, the real-time QoE in each scheduling slot is depicted in Fig. 4(a). Initially, the QoE value remains low as the user DT model inference conducts multicast grouping and updates the swipe probability distribution, which pauses the video transcoding and transmission process. Around step 20, QoE decreases significantly because the current user DT model cannot capture new swipe behavior patterns. Then, QoE slightly improves but is disrupted again around step 90 due to changes in swipe behaviors, necessitating another user DT model update. Overall, the adaptive user DT model inference and update scheme can maintain user QoE at a relatively high and robust level.

For the performance comparison, we select two distinct schemes. Specifically, the fixed user DT scheme employs the fixed user DT model inference version and update frequency, maintaining the same 3C management strategy. The hierarchical DRL scheme utilizes the same adaptive user DT model inference and update strategy, but its 3C management strategy is based on a two-layer reward-driven DRL algorithm. As shown in Fig. 4(b), as bandwidth resources increase, user QoE gradually improves, although the margin of improvement diminishes. This indicates that there is an upper limit to the enhancement of QoE with additional network resources. When the bandwidth reaches 2.2 MHz, the average QoE of the proposed scheme is 10.3% higher than that of the hierarchical DRL scheme. Overall, the adaptive user DT model inference and update scheme with generative AI-based 3C management can achieve high QoE performance.

¹ YouTube 8M dataset: <https://research.google.com/youtube8m/index.html>

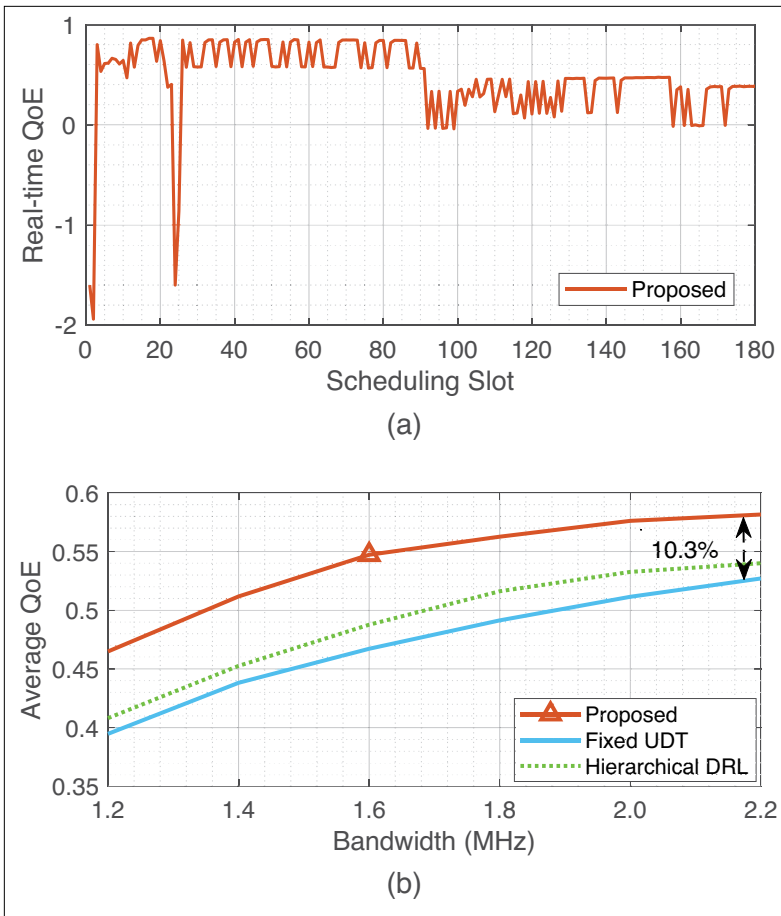


FIGURE 4. The QoE performance comparison. a) Real-time QoE performance in different scheduling slots. b) Average QoE performance with respect to the increase of bandwidth.

OPEN RESEARCH ISSUES

HIERARCHICAL NETWORK DT DEPLOYMENT

User, infrastructure, and slice DTs correspond to different network entities, typically deployed at various levels of network nodes: low-level, medium-level, and high-level. However, due to the dynamic and heterogeneous nature of communication networks, it is important to explore how to flexibly deploy network DTs to enhance network performance. For instance, in scenarios with high mobility where users frequently switch between base stations, deploying user DTs at medium-level nodes can ensure service continuity. Furthermore, to achieve load balancing between macro base stations, infrastructure DTs can be strategically positioned at high-level nodes. Similarly, slice DTs can be flexibly deployed at medium-level and high-level nodes to maintain high service quality across different service areas.

HYBRID DATA-MODEL DRIVEN DECISION-MAKING

Hybrid data-model driven decision-making in the network DT integrates data-driven and model-driven approaches to effectively tackle complex network management problems. Purely data-driven AI methods often struggle to achieve optimal solutions and may lack stability, particularly when confronted with unfamiliar scenarios.

By decoupling the network management problem, convex sub-problems are solved using model-based techniques for precision and reliability, while complex non-convex sub-problems are addressed with data-driven methods to manage their intricacies. Through iterative interaction between these methods, the network DT decision-making progressively converges toward the optimal solution.

EFFICIENT NETWORK DT COLLABORATION

Given the complex tasks that frequently arise in communication networks, such as cooperative sensing and collaborative inference, individual network DTs struggle to enhance the collaborative capabilities of communication networks. One promising area of research is the integration of advanced multi-agent system algorithms with network DTs. This approach would enable each network DT to function as an independent agent that can negotiate, coordinate, and cooperate with other network DTs in real-time. Moreover, a federated learning framework, where multiple network DTs share insights without exchanging raw data, can preserve privacy and facilitate predictive maintenance across different network segments [15].

SECURE AI AND DT INTEGRATION

The integration of AI and DT technologies into 6G network management introduces significant security challenges that need further investigation. On the one hand, AI models embedded in DTs are vulnerable to adversarial attacks, data poisoning, and model inversion, which may compromise the reliability of network status prediction and decision-making.

On the other hand, DTs continuously collect and synchronize vast amounts of user and network data, raising concerns over data privacy, integrity, and secure transmission. Hence, future research should focus on developing security-aware AI and DT mechanisms, such as privacy-preserving data sharing frameworks and trustworthy model validation schemes, to ensure safe and reliable deployment of AI-native network DTs.

CONCLUSION

We propose a novel AI-native network DT framework to facilitate intelligent network management for 6G. Advanced AI models establish network DT models and collaborate to perform intricate network analyses and make decisions. Network DTs interact with the physical network to adjust the data collection frequency for timely updates of the DT models. This proposed framework enables efficient and customized management of holistic mobile communication networks to enhance user QoE. A case study is presented, and several open research issues are discussed to accelerate the development of network DTs.

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